C# Machine Problem: Loan Eligibility and Risk Assessment System

Problem scenario

A financial institution needs a console-based application that evaluates a customer's eligibility for a personal loan. The system must analyze the applicant's age, monthly income, employment status, credit score, existing debt, and requested loan amount before determining the final decision.

Your task is to develop a Smart Loan Eligibility and Risk Assessment System using only if, else if, else, and nested conditional statements.

Program requirements

The program must collect the following information from the applicant:

Input	Description
Age	Applicant's age in years
Monthly income	Gross monthly income in pesos
Employment status	Employed, self-employed, or unemployed
Years employed	Number of years in current employment or business
Credit score	Score from 300 to 850
Existing monthly debt	Total monthly debt payments in pesos
Requested loan	Amount of money being borrowed

Decision-making rules

Students must implement the following rules in the correct order.

1. Initial eligibility

- Applicants younger than 18 are automatically rejected.
- Applicants aged 18–20 may qualify only if their monthly income is at least ₱50,000 and their credit score is at least 750.
- Applicants older than 65 must undergo manual review.
- Applicants with a monthly income below ₱15,000 are rejected.

2. Employment assessment

- Employed: Must have at least 1 year of employment.
- Self-employed: Must have at least 2 years in business.
- Unemployed: Automatically rejected.
- Applicants with insufficient employment history must undergo manual review if their monthly income is at least ₱40,000; otherwise, they are rejected.

3. Credit risk classification

Credit score	Classification
750–850	Excellent
700–749	Good
650–699	Fair
600–649	Poor
Below 600	Very poor

Rules:

- Scores below 600 result in rejection.
- Scores from 600–649 require manual review.
- Scores of 650 or higher may proceed to the next assessment.

4. Debt-to-income ratio (DTI)

Calculate the debt-to-income ratio using:

$$\text{DTI}=\frac{\text{Existing Monthly Debt}}{\text{Monthly Income}}\times100$$

- DTI of 30% or less: Low risk.
- DTI above 30% up to 40%: Moderate risk.
- DTI above 40% up to 50%: High risk; manual review required.
- DTI above 50%: Reject the application.

5. Loan amount assessment

Calculate the requested loan as a multiple of monthly income.

Loan amount relative to monthly income	Decision
Up to 5 times	Eligible for standard assessment
Above 5 times up to 10 times	Requires good credit (700+) and DTI of 30% or less
Above 10 times	Reject the application

6. Final decision

The program must display exactly one final decision:

- LOAN APPROVED
- FOR MANUAL REVIEW
- LOAN REJECTED

If multiple conditions apply, rejection takes priority over manual review, and manual review takes priority over approval.